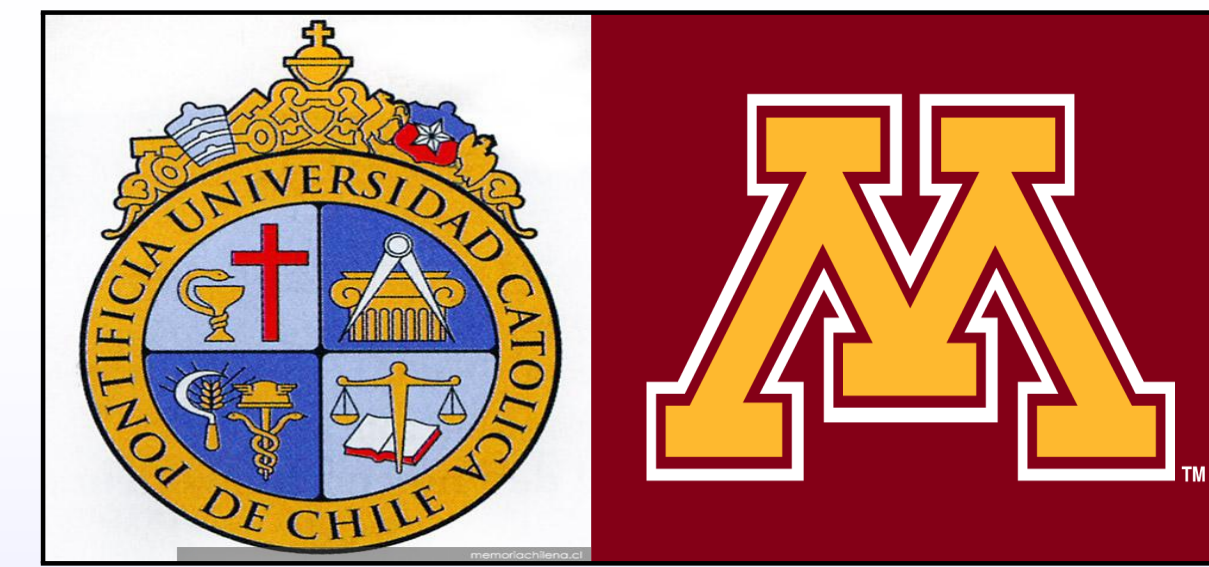


Hybridizable Discontinuous Galerkin Methods for the 2D Vector Laplacian Equation

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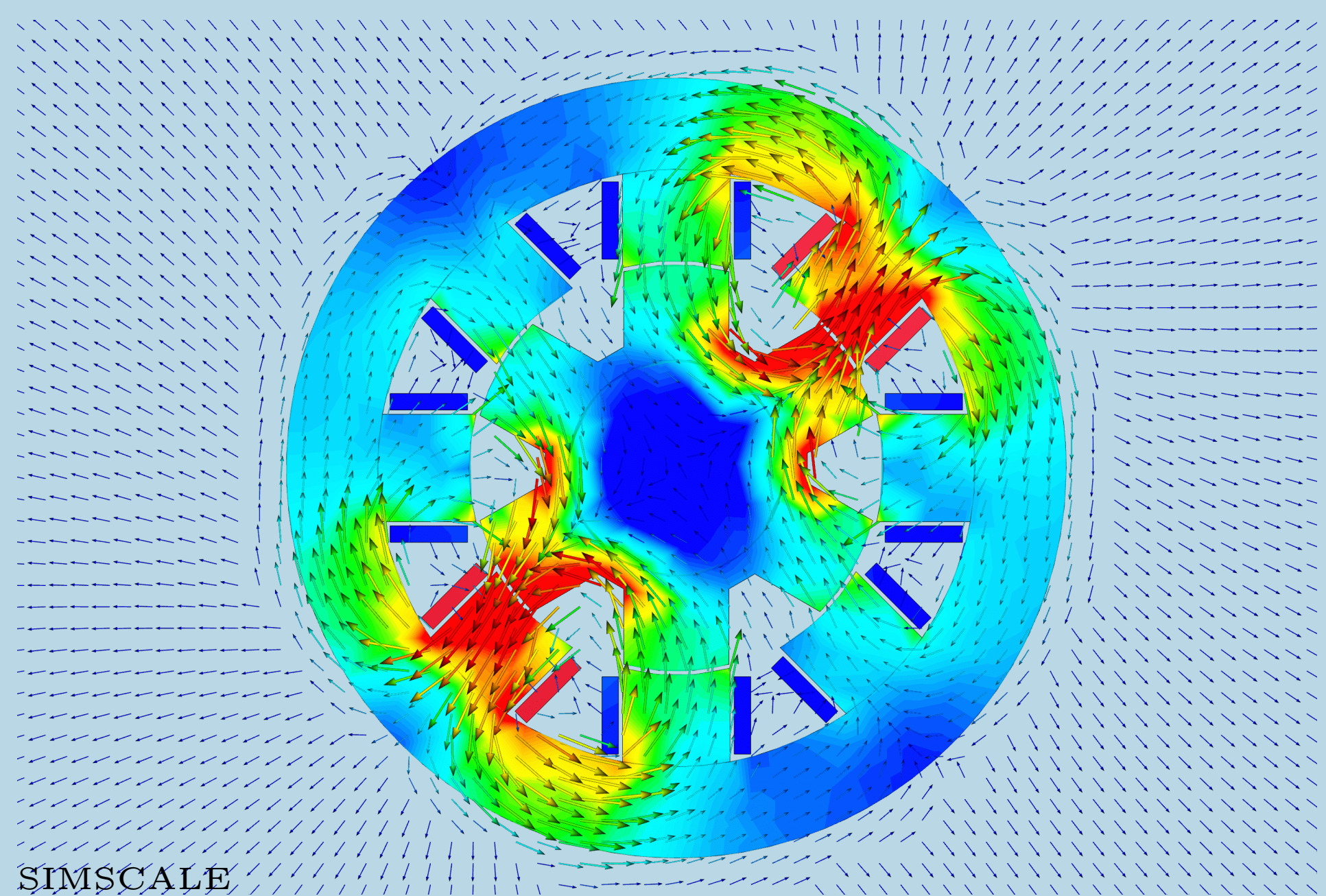


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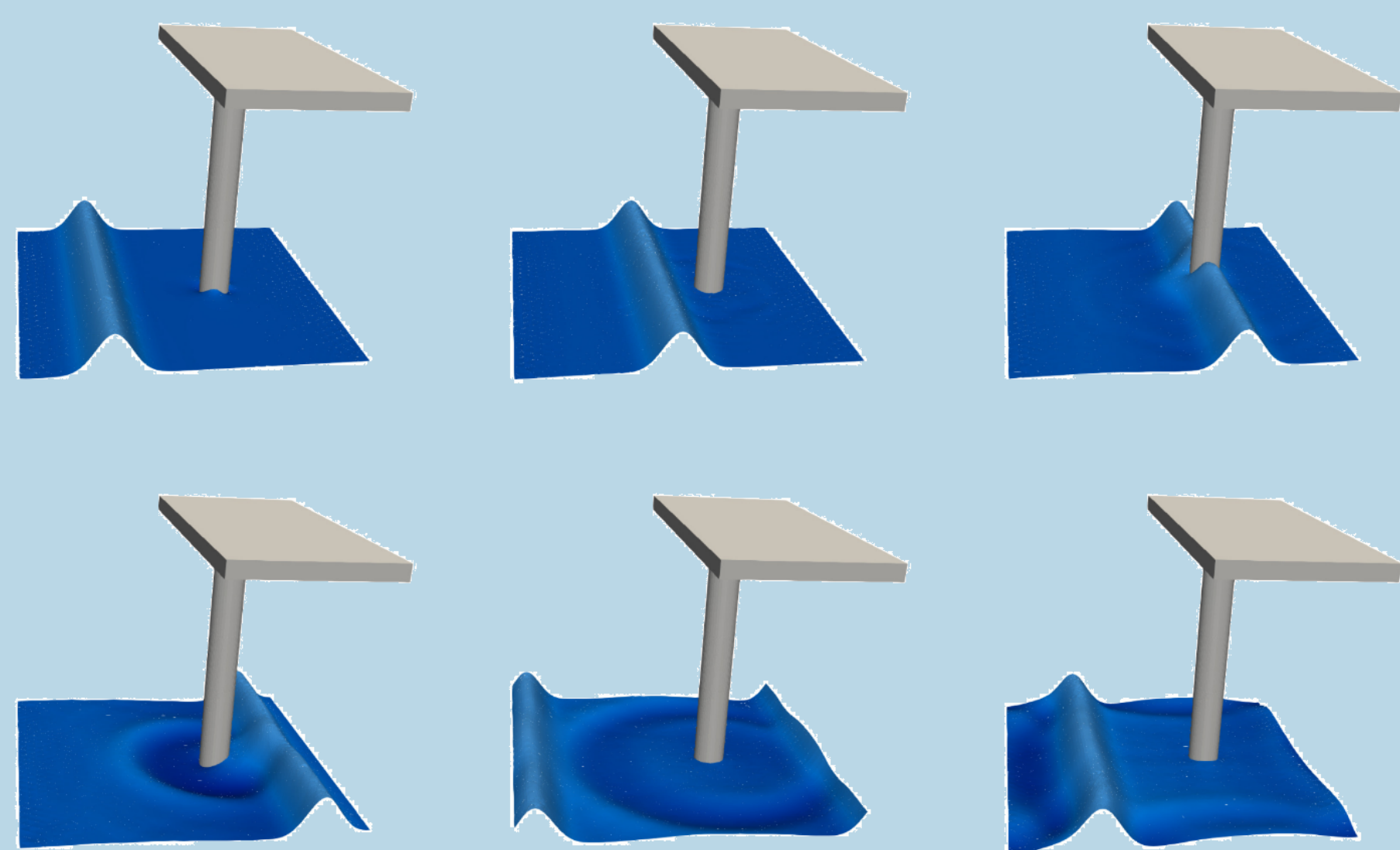
Problem

Given $\mathbf{f} \in \mathbf{L}^2(\Omega)$ with $\Omega \subset \mathbb{R}^2$ a bounded Lipschitz domain, find $\mathbf{u}$ such that

$$\mathbf{curl} \, \mathbf{rot} \, \mathbf{u} - \mathbf{grad} \, \mathbf{div} \, \mathbf{u} = \mathbf{f}, \quad \text{in } \Omega.$$



SIMSCALE



Applications

- 1. Electromagnetics:** magnetic field or electric field.
- 2. Fluid Dynamics:** velocity, displacement, vorticity.

Mixed Formulation

The **rotational-divergence mixed** formulation is

$$\sigma - \mathbf{rot} \, \mathbf{u} = 0, \quad \text{in } \Omega,$$

$$\phi + \mathbf{div} \, \mathbf{u} = 0, \quad \text{in } \Omega,$$

$$\mathbf{curl} \, \sigma + \mathbf{grad} \, \phi = \mathbf{f}, \quad \text{in } \Omega,$$

Boundary conditions:

$\sigma = 0$	$\mathbf{u} \cdot \mathbf{n}^\perp = 0$	
$\phi = 0$	Type I	Type II
$\mathbf{u} \cdot \mathbf{n} = 0$	Type III	Type IV

References

- [1] Bernardo Cockburn and Jayadeep Gopalakrishnan, *The derivation of hybridizable discontinuous Galerkin methods for Stokes flow*, SIAM J. Numer. Anal., 47(2):1092–1125, 2009.
- [2] Douglas N. Arnold, Richard S. Falk, and Jay Gopalakrishnan, *Mixed finite element approximation of the vector Laplacian with Dirichlet boundary conditions*, Math. Models Methods Appl. Sci., 22(9):1250024, 26, 2012.
- [3] Gerard Awanou, Maurice Fabien, Johnny Guzmán, and Ari Stern, *Hybridization and postprocessing in finite element exterior calculus*, Math. Comp., 92(339):79–115, 2023.

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The HDG Method

Find $(\sigma_h, \phi_h, \mathbf{u}_h) \in \mathbf{V}_h \times W_h \times W_h$ such that

$$(\sigma_h, \chi)_{\mathcal{T}_h} - (\mathbf{u}_h, \mathbf{curl} \, \chi)_{\mathcal{T}_h} + \langle \tilde{\mathbf{u}}_h \cdot \mathbf{n}^\perp, \chi \rangle_{\partial \mathcal{T}_h} = 0,$$

$$(\phi_h, \varphi)_{\mathcal{T}_h} - (\mathbf{u}_h, \mathbf{grad} \, \varphi)_{\mathcal{T}_h} + \langle \hat{\mathbf{u}}_h \cdot \mathbf{n}, \varphi \rangle_{\partial \mathcal{T}_h} = 0,$$

$$(\sigma_h, \mathbf{rot} \, \mathbf{z})_{\mathcal{T}_h} + \langle \check{\sigma}_h, \mathbf{z} \cdot \mathbf{n}^\perp \rangle_{\partial \mathcal{T}_h} - (\phi_h, \mathbf{div} \, \mathbf{z})_{\mathcal{T}_h} + \langle \hat{\phi}_h, \mathbf{z} \cdot \mathbf{n} \rangle_{\partial \mathcal{T}_h} = (\mathbf{f}, \mathbf{z})_{\mathcal{T}_h},$$

for all $(\chi, \varphi_h, \mathbf{z}) \in \mathbf{V}_h \times W_h \times W_h$.

The **numerical fluxes** are:

$$\tilde{\mathbf{u}}_h \cdot \mathbf{n}^\perp = \mathbf{u}_h \cdot \mathbf{n}^\perp + \alpha(\sigma_h - \check{\sigma}_h),$$

$$\hat{\mathbf{u}}_h \cdot \mathbf{n} = \mathbf{u}_h \cdot \mathbf{n} + \tau(\phi_h - \hat{\phi}_h).$$

Implementations

Method	Global variables	Global problem
HDG I	$\check{\sigma}_h \in M_h$ $\hat{\phi}_h \in M_h$	$\langle \tilde{\mathbf{u}}_h \cdot \mathbf{n}^\perp, \mu \rangle_{\partial \mathcal{T}_h \setminus \Gamma} = 0, \quad \forall \mu \in M_h$ $\langle \hat{\mathbf{u}}_h \cdot \mathbf{n}, \mu \rangle_{\partial \mathcal{T}_h \setminus \Gamma} = 0, \quad \forall \mu \in M_h$
HDG II	$\tilde{\mathbf{u}}_h \in \mathbf{N}_h^\perp$ $\hat{\phi}_h \in M_h$	$\langle \check{\sigma}_h, \boldsymbol{\mu} \cdot \mathbf{n}^\perp \rangle_{\partial \mathcal{T}_h \setminus \Gamma} = 0, \quad \forall \boldsymbol{\mu} \in \mathbf{N}_h^\perp$ $\langle \hat{\mathbf{u}}_h \cdot \mathbf{n}, \mu \rangle_{\partial \mathcal{T}_h \setminus \Gamma} = 0, \quad \forall \mu \in M_h$
HDG III	$\check{\sigma}_h \in M_h$ $\hat{\mathbf{u}}_h \in \mathbf{N}_h$	$\langle \tilde{\mathbf{u}}_h \cdot \mathbf{n}^\perp, \mu \rangle_{\partial \mathcal{T}_h \setminus \Gamma} = 0, \quad \forall \mu \in M_h$ $\langle \hat{\phi}_h, \boldsymbol{\mu} \cdot \mathbf{n} \rangle_{\partial \mathcal{T}_h \setminus \Gamma} = 0, \quad \forall \boldsymbol{\mu} \in \mathbf{N}_h$
HDG IV	$\tilde{\mathbf{u}}_h \in \mathbf{N}_h^\perp$ $\hat{\mathbf{u}}_h \in \mathbf{N}_h$	$\langle \check{\sigma}_h, \boldsymbol{\mu} \cdot \mathbf{n}^\perp \rangle_{\partial \mathcal{T}_h \setminus \Gamma} = 0, \quad \forall \boldsymbol{\mu} \in \mathbf{N}_h^\perp$ $\langle \hat{\phi}_h, \boldsymbol{\mu} \cdot \mathbf{n} \rangle_{\partial \mathcal{T}_h \setminus \Gamma} = 0, \quad \forall \boldsymbol{\mu} \in \mathbf{N}_h$

A Priori Error Estimates

Theorem. *There exist C and $\bar{C}$ such that*

$$\|\sigma - \sigma_h\|_{\mathcal{T}_h} + \|\phi - \phi_h\|_{\mathcal{T}_h} \leq C(h^{k+\frac{1}{2}}\|\mathbf{u}\|_{\mathbf{H}^s(\Omega)} + h^{k+\frac{1}{2}}\|\sigma\|_{H^t(\Omega)} + h^{k+\frac{1}{2}}\|\phi\|_{H^r(\Omega)}),$$

$$\|\mathbf{u} - \mathbf{u}_h\|_{\mathcal{T}_h} \leq \bar{C}(h^{k+1}\|\mathbf{u}\|_{\mathbf{H}^s(\Omega)} + h^{k+1}\|\sigma\|_{H^t(\Omega)} + h^{k+1}\|\phi\|_{H^r(\Omega)})$$

Numerical experiments:

